EIU

Assignment 5: Creating a VPC with Subnets

Overview

This assignment provides a hands-on experience in designing and implementing a secure, highly available, and scalable network infrastructure using Amazon Virtual Private Cloud (VPC). You will build a multi-tier, multi-AZ network from the ground up, configure routing for public and private subnets, implement layered security controls, and establish private connectivity between VPCs and to AWS services. The objective is to develop a deep, practical understanding of core AWS networking components and best practices. This assignment was inspired by the work mentioned in Chapter 3 of the textbook.

Deliverables

Each student must submit the following:

1. **Presentation Slides (15–30 slides maximum)**
    - Title slide with student name(s), course, and assignment title.
    - Clear explanation of each completed task: (Access AWS academy: Module 5 Networking and Content Delivery - Lab 2: Build your VPC and Launch a Web Server)
        1. Task 1- Create a VPC.
        2. Task 2 - Create subnets.
        3. Task 3 - Configure a security group.
        4. Task 4 - Launch an EC2 instance into a VPC.
        5. Task 5 - Creating a VPC Endpoint for S3 (Follow Lesson - 4 PPT (Create a VPC endpoint for S3 in the us-east1 region instruction)
        6. Task 6 – Answering Reflection Questions
    - Reflection question responses (**Choose any three (3). M**ust be included in slides with appropriate answer). Provide clear, concise, and well-reasoned answers in your presentation.
        1. What is the key difference between an Internet Gateway and a NAT Gateway? Explain the specific purpose each one served in this assignment's architecture.
        2. Explain the difference between a Security Group (stateful) and a Network ACL (stateless). Why is it a best practice to use both as a "defense in depth" strategy?
        3. Why must a NAT Gateway be placed in a public subnet, and what other two components did it depend on to function correctly?
        4. If you add a new private subnet to your VPC (e.g., for a new service), what is the minimum number of steps or resources (like route table associations) required to grant its instances outbound internet access?
        5. What would happen if you tried to access S3 from the private instance without the endpoint or a NAT Gateway?
        6. This assignment used two Availability Zones for high availability. How does placing subnets in a second AZ protect your application? What would be the single point of failure if your entire VPC (all subnets) was in a single AZ?

- Screenshots of key steps / results with brief description of each steps / results.
- Lessons learned and challenges encountered.

Presentation Guidelines

- Use professional formatting (legible fonts, consistent colors, organized layout).
- Ensure clarity: avoid overcrowding slides with text; use bullet points and visuals.
- Provide screenshots of completed task as evidence of work completed.
- Answer about the process and challenges you suffer while performing each task.

1. Practical Task Completion (60 points total)

Each AWS task is evaluated based on the correctness, completeness, and evidence (e.g., each task results solved screenshots should be enclosed).

2. Presentation Quality (30 points total)

- Clear structure (intro, results, discussion, conclusion)
- Slides visually organized and professional, for example, adding appropriate title of each task, pictures, transitions, animations Screenshots, included as evidence of each task results with proper description of each task results.

3. Professionalism & Delivery (10 points total)

- Confident and clear presentation delivery
- Submit slides (PowerPoint or PDF) to Moodle by the deadline.

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